

TUNKA, Russian Federation

WMO: 308110

Lat: 51.733N

Lon: 102.533E

Elev: 721

StdP: 92.96

Time Zone: 8.00 (E08)

Period: 82-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.6	(c) -34.4	(d) -38.6	(e) 0.1	(f) -36.5	(g) -36.5	(h) 0.1	(i) -34.3	(j) 6.9	(k) -16.7	(l) 5.8	(m) -14.3	(n) 0.0	(o) 330	(p) 0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	27.2	17.0	25.3	16.3	23.5	15.5	18.4	24.7	17.4	23.3	16.5	21.9	2.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.6	20.3	15.2	11.8	19.5	14.3	11.1	18.7	55.3	24.8	52.1	23.7	49.1	22.2	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.3	(b) 6.2	(c) 5.1	(e) -39.0	(f) 30.4	(g) 2.5	(h) 2.0	(i) -40.8	(j) 31.9	(k) -42.2	(l) 33.0	(m) -43.6	(n) 34.2	(o) -45.4	(p) 35.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.7	-24.5	-20.3	-8.6	2.0	9.3	14.4	17.0	15.3	8.5	0.3	-12.4	-22.1
	DBStd	15.39	5.05	5.56	6.52	4.15	4.02	3.35	2.61	2.78	3.55	4.43	6.30	5.54
	HDD10.0	4842	1069	848	577	240	63	4	1	2	70	302	673	994
	HDD18.3	7324	1328	1081	835	489	282	123	56	99	296	560	923	1253
	CDD10.0	588	0	0	0	1	41	136	218	166	24	1	0	0
	CDD18.3	27	0	0	0	0	1	6	15	6	0	0	0	0
	CDH23.3	419	0	0	0	0	47	109	181	81	2	0	0	0
	CDH26.7	74	0	0	0	0	8	22	37	8	0	0	0	0
Wind	WSAvg	1.2	0.4	0.6	1.2	2.2	2.1	1.9	1.4	1.3	1.4	1.1	0.7	0.4
Precipitation	PrecAvg	371	4	3	4	13	26	59	101	87	48	13	7	4
	PrecMax	515	9	12	11	39	58	108	189	150	100	44	21	16
	PrecMin	249	0	0	0	2	6	21	26	29	16	1	0	0
	PrecStd	60	3	3	3	9	13	23	36	28	21	9	4	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.7	-0.5	11.8	20.2	27.9	29.3	29.9	27.9	22.5	15.9	4.3	-4.8
		MCWB	-7.9	-3.0	4.7	9.9	15.0	17.4	18.9	18.1	13.9	8.2	0.7	-6.1
	2%	DB	-11.0	-5.7	6.9	15.9	24.5	26.3	27.4	25.5	19.7	12.5	1.3	-8.4
		MCWB	-11.7	-7.2	1.8	7.6	13.2	15.7	17.9	17.2	12.2	6.3	-1.0	-9.3
	5%	DB	-13.8	-8.2	3.9	13.2	21.6	24.2	25.3	23.5	17.4	9.7	-1.3	-11.2
		MCWB	-14.4	-9.4	0.0	5.6	11.4	15.0	17.3	16.3	11.2	4.9	-2.9	-11.8
	10%	DB	-16.4	-10.7	1.4	10.4	18.8	22.0	23.4	21.7	15.2	7.3	-3.3	-13.8
		MCWB	-16.8	-11.7	-1.6	4.1	10.2	14.0	16.4	15.4	10.2	3.5	-4.7	-14.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.5	-3.1	5.0	10.3	15.9	18.5	20.1	19.4	14.9	8.7	1.0	-6.2
		MCDB	-6.4	-0.7	11.4	18.6	26.1	26.5	27.1	25.6	20.2	15.1	3.9	-5.1
	2%	WB	-11.8	-7.2	2.0	7.7	13.7	17.0	18.9	17.8	13.1	6.6	-0.9	-9.3
		MCDB	-11.1	-5.7	6.4	15.6	23.4	23.9	25.4	23.8	18.4	11.9	1.2	-8.4
	5%	WB	-14.4	-9.4	0.2	6.0	12.1	15.9	17.9	16.8	11.8	5.1	-2.7	-11.8
		MCDB	-13.8	-8.2	3.7	12.6	20.3	22.4	23.6	22.4	16.5	9.2	-1.3	-11.2
	10%	WB	-16.8	-11.6	-1.5	4.3	10.6	14.8	17.0	15.9	10.6	3.7	-4.6	-14.3
		MCDB	-16.4	-10.6	1.4	10.0	17.9	20.8	22.4	20.6	14.7	7.0	-3.4	-13.8
Mean Daily Temperature Range	5% DB	MDBR	11.1	14.7	14.5	12.7	14.0	11.8	11.0	11.1	12.3	11.6	10.8	9.8
		MCDBR	12.9	16.3	15.8	18.9	20.7	17.3	15.6	15.4	16.5	16.4	11.9	12.1
		MCWBR	12.2	14.6	10.7	10.9	10.6	7.8	6.9	7.2	8.9	9.9	9.6	11.4
	5% WB	MCDBR	12.9	16.4	15.4	18.3	18.7	15.0	13.4	13.5	14.6	15.3	11.9	12.3
		MCWBR	12.2	14.5	10.4	10.7	9.8	7.1	6.8	6.6	8.0	9.4	9.6	11.5
Clear-Sky Solar Irradiance	taub		0.248	0.276	0.319	0.364	0.408	0.402	0.399	0.368	0.321	0.300	0.274	0.251
	taud		2.223	2.193	2.167	2.229	2.168	2.310	2.362	2.425	2.499	2.454	2.292	2.183
	Ebn at Noon		770	845	869	871	844	853	849	860	871	819	727	688
	Edh at Noon		66	92	117	125	140	123	115	102	84	72	62	58
All-Sky Solar Radiation	RadAvg		1.35	2.43	3.85	4.92	5.40	5.51	5.12	4.46	3.55	2.32	1.39	0.98
	RadStd		0.06	0.10	0.18	0.23	0.26	0.34	0.39	0.28	0.30	0.13	0.08	0.05

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page